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STUDIES IN FLUIDIZED BED COMBUSTOR OF 7.5MW PLANT BASED ON COTTON STALKS

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Abstract

The use of biomass fuels for energy production through combustion has a growing application worldwide mainly for two reasons: first, the utilization of biomass for energy contributes to mitigate emission of green house gases; second, its use decreases the dependence of imported fossil fuels in India. Fluidized-bed combustion process seems to be suitable technology for converting it into thermal energy and power generation. Fluidized-bed combustion (FBC) is one of the most promising energy conversion options available today. FBC combines high efficiency combustion of low-grade fuels viz. high ash coal, coal washery rejects and middling, wood and other biomass of agri-waste and municipality waste. Cotton stalks are found abundantly in Bathinda-Muktsar belt of Punjab, as cotton crop is more suitable to this area. These cotton stalks could generate huge amount of power by incinerating in fluidized bed combustors. In this work, a mathematical model has been developed for physico chemical parameters of 7.5 MW AFBC power plant using cotton stalks. First of all description of above plant is discussed. There is wide variation in size of cotton stalks which leads us to do sieve analysis in lab to calculate the average size of cotton stalks that are fed into fluidized bed combustor at plant. Physico-chemical parameters of plant have been calculated using physico chemical model and parameters taken from plant. Finally Thermogravimetric analysis has been done on powdered cotton stalks on recently installed Perkin Elmer-6 in Mechanical Engineering Department, Thapar University.

Key words: Cotton stalks, physico- chemical model and combustor

1.0 Introduction

In the past 20 years biomass has become more attractive to industry since it is a low CO₂, alternative energy source. Due to its easy availability biomass is still the most traditional fuel used in the rural sector all over the world, where it is utilized in a highly inefficient manner with a low overall efficiency, contributing significantly to environmental pollution and health hazards. Cotton stalks are found abundantly in Bathinda-Muktsar belt of Punjab, as cotton crop is more suitable to this area. These cotton stalks could generate huge amount of power by incinerating in fluidized bed combustors. Sun et al. (2008) presented work reports studies on the mixing and combustion characteristics of cotton stalk (CS) with 10–100 mm in length in a fluidized bed. CS can mix well with 0.6–1 mm alumina at fluidization number $N=3-8$. Fluidized bed combustion of pure CS was conducted in a 0.2 MW test facility. The concentrations of major gaseous pollutants (CO, SO₂ and NO) in flue (stack) gas were measured. The combustion efficiency of CS is over 99.5%. Additionally, the experiments researched the agglomeration characteristics of bed materials: silica sand and alumina. SEM/EDX analyses on silica samples suggest that the high alkali metals content may cause the formation of the coating around silica particles. The coating consists of compounds with low melting temperature and can result in the bed agglomeration. By contrast, alumina is difficult to react with alkali metals from

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CS ash after 38 h combustion. R.P. Liu et al. (2007) did comparison of two kinds of bed materials during CFB combustion of cotton stalks. Silica and aluminous soil were used as bed materials for the combustion of cotton stalk in a Circulating Fluidized Bed (CFB) combustion pilot plant (0.2 MW). After a long period of operation, the silica bed material was found to have sintered and the aluminous soil bed material was found to have resisted sintering. The particles of both types of bed materials were sampled. Two instrumental approaches (XRF and SEM/EDS) were employed in order to identify structural reasons for the observed differences in sintering behavior between the two bed materials when they were used in the same experimental conditions. From these experiments it was proven that the aluminous soil particles eliminate the bed agglomeration. Although the aluminous soil particles were enriched with alkali metals and alkaline-earth metals after burning for 38 h, they still did not form any agglomerates. Therefore, these particles should provide a good bed material to substitute for the traditional silica bed materials in the CFB combustion of biomass with alkali ashes. In this work, a mathematical model has been developed for calculating physico chemical parameters for 7.5 MW FBC power plant using cotton stalks. Also the description of above plant is discussed. TGA of powdered cotton stalks has been done to correlate it with combustion properties. Also techno economic analysis of above plant has been done has been

2.0 Description of 7.5 MW Malwa power plant

Malwa power limited, Muktsar, a 7.5 MW FBC based power plant is the plant from which the data for the present work was collected. It is located in the south-western district of Punjab, named Muktsar. This area of Punjab is popularly called cotton belt of Punjab. As the soil is quite conducive for cotton, so the farmers of south-western Punjab prefer growing cotton. Above all, many farmers grow cotton twice a year. A large amount of cotton waste is left in the fields after the cotton has been harvested from the plants. This waste is called 'cotton stalk' and has a height varying from 3 feet to 5 feet. Cotton stalk is quite rich in carbon content. Traditionally, this was used for domestic cooking and water heating purposes; this caused a lot of pollution. Malwa power plant was setup in 2005; primarily it is based upon the combustion of cotton stalks. The plant collects the cotton stalk directly from the farmers and stores it in the vast land area acquired by it. First of all the collected and stored cotton stalk is passed through a wood chipper for breaking the stalk into appropriate size for the combustor. The usual size to which the stalk is cut, is around 2.5 cm to 4 cm. Figure 1 shows the process flow of the above power plant.

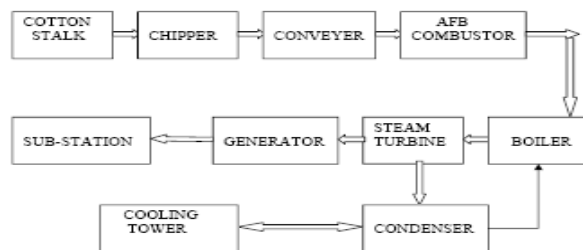


Figure 1 Process flow of the power plant

Then the chipped material is moved with the help of semi-automatic trucks and heaps of it are prepared. The chipped fuel is manually loaded onto a belt conveyor. The belt conveyor drops the material into two compartments fitted with screw feeders, which then supply the fuel to the fluidized bed combustor. Before running the combustor on cotton stalk, it is fired with the help oil and coal. When the combustor reaches a temperature of around 700°C, then the cotton stalk is continuously supplied. The boiler is supplied with the filtered and treated water from water treatment plant. Superheated steam at temperature of 475°C and around 67 bar pressure is produced in the boiler and is stored in the steam drum before passing on to the turbine. The turbine is coupled with the generator, which produces power. The generated power is stepped up, synchronized and is transmitted to the nearest 32KV sub-station of the electricity board. Table 1 shows the plants parameters. Figure 2 shows photograph of chipper at the *International Conference on 'Advances in Energy Research', (ICAER) 2009*

plant site. Figure 3 shows the actual photograph of boiler at the plant site. Figure 4 shows the actual photograph of cotton stalks that fed into the fbc boiler.

Table 1 Plant parameters

Capacity	7.5MW	Distribution plate	Nozzle type
Auxiliary consumption	11.5%	Bed material	50% alumina
Type of Bed	AFBC	Fuel used	Cotton stalk
Bed Temeperature	750-800 ⁰ C	Fly-ash collector	ESP
Bed cross-sectional area	19.43 m ²	Fuel feed rate	10 T/hr
Steam temperature	475°C	Exit gas temperature	150°C
Feed water temperature	100°C	Steam production rate	34 T/hr



Figure 2. Chipper in progress



Figure 3. FBC boiler



Figure 4. Sample of cotton stalk

3.1 Mathematical Model of Atmospheric Bubbling Fluidized Bed Combustor

A mathematical model is formed to describe the physico-chemical properties of fluidized bed biomass combustor, is shown below. The input data for the model has been proximate analysis and physico chemical parameters taken from plant.

3.1 Superficial Velocity

The stoichiometric air feed rate is determined from simple stoichiometrical relationship as follows;

$$F_{MTH} = W_{fuel} (1.0 - XW) \left[\frac{XC}{12.0} + \frac{XH}{4.0} + \frac{XS}{32.0} - \frac{XO}{32.0} \right] \frac{1}{0.21} \quad (1)$$

Where W_{fuel} is the fuel feed rate into the combustor (g/s), and XC , XH , XS , XO , and XW are the ultimate carbon, hydrogen, sulphur, oxygen and moisture content of the coal respectively. The actual molar feed rate of fluidizing air is calculated as; Where E_{xair} fractional excess air

Now the superficial velocity is calculated as;

$$F_{ME} = F_{MTH} (1.0 + E_{xair}) \text{ g-mole/s} \quad (2)$$

$$U_0 = \frac{F_{ME} R_g T_b}{P_{av} A_t} \text{ cm/s} \quad (3)$$

Where $R_g = 82.06 \text{ atm-cm}^3/(\text{g-mole})(\text{K})$, P_{av} : Average pressure in the combustor, T_b : Absolute bed temperature (K) and A_t : Cross-sectional area of the bed (cm^2)

3.2 Minimum Fluidization Velocity

The Wen and Yu (1966) correlation was used to calculate the minimum fluidization velocity in the present model because the particles are greater than $100 \mu\text{m}$.

$$U_{mf} = \left(\frac{\mu_g}{d_p \cdot \rho_g} \right) \left[\left(33.7 \right)^2 + \frac{0.0408 d_{pav} \rho_g (\rho_p - \rho_g) g}{\mu_g^2} \right]^{0.3} - 33.7 \text{ cm/s} \quad (4)$$

3.3 Viscosity of Fluidizing Gas

The final viscosity expression, which is a function of bed temperature only, is presented below;

$$\mu_g = 1.4 (10^{-5}) (T_b)^{1/2} \text{ g/cm.s} \quad (5)$$

3.4 Average Equivalent Bubble Diameter

The correlation proposed by Mori and Wen (1975) taking into account the effect of bed diameter and distributor type on bubble diameter is used here.

$$D_b = 0.43 (U_o - U_{mf})^{0.4} (Z + 4(A_o)^{0.5})^{0.8} \text{ g}^{-0.2} \quad (6)$$

Where D_b is Diameter of Bubble, A_o is Bed Area and Z is the minimum fluidization height

3.5 Reaction Rate Constant

In this case, the single film theory of char combustion for shrinking particles has been considered where carbon reacts with oxygen to produce CO_2 . Since bed temperature is below 1373 K . The surface reaction rate constant K_s for carbon particles, is adopted from the expression given by Parker and Hottel(1936) which is as follows

$$K_s = 4.32 \times 10^{11} T_p^{-0.5} \cdot \exp(-44000 / RT_p) \quad (7)$$

The mass transfer coefficient K_g is expressed by the dimensionless Sherwood number (Sh) as

$$K_g = \frac{Sh \cdot D_g}{d_p} \quad (8)$$

Therefore the chemical reaction rate constant (K) at any instant, which is controlled by surface reaction and gas diffusion coefficients is given as:

$$1/K = 1/K_s + 1/K_g \quad (9)$$

For estimating the Sherwood number the La Nauze et al.(1984) correlation has been adopted, which was developed on the assumption of frequent renewal of gas at the particle surface and correlation is given as:

$$Sh = 2\varepsilon_{mf} + \left[4.0d_{pi} \left(\frac{U_{mf}}{\varepsilon_{mf}} + U_b \right) / \pi D_g \right]^{0.5} \quad \text{for } d_p / d_{pbed} > 4.0 \quad (10)$$

$$Sh = 2\varepsilon_{mf} + \left[4.0d_{pi} U_{mf} / \pi D_g \right]^{0.5} \quad \text{for } d_p / d_{pbed} \leq 4.0 \quad (11)$$

4.0 Results and Discussion for model developed.

The input data required for the model development has been taken from the Malwa power Ltd., Muktsar (Punjab). The plant is a 7.5 MW thermal power plant using the FBC process. The input data is proximate and ultimate analysis of cotton stalks samples. The result of proximate analysis is moisture 13.89 %, ash 4.0%, volatile matter 60.5% fixed carbon 21.61% and ultimate analysis is C 42.37 %,H 5.66 %,S 0.00 %,O 36.58 % and N 1.50%. The average size of cotton stalks that are fed in combustor is 2 to 4.5 cm. The various parameters used for calculation are given in table 1. The average size of cotton stalks is assumed in model as 3 cm. The results of model are :Superficial velocity- 222.3603 cm/s, Inlet molar oxygen input- 2.5018×10^{-6} g-mole/cm³, Viscosity of gas used- 4.4778×10^{-4} g/cm.s, Density of gas used- 3.4526×10^{-4} g/cm³, Minimum fluidization velocity required-186.4129 cm/s, Bubble diameter- 13.5 cm, K_s = 167.6369 cm/s, Sh_{no} = 22.3524, K_g = 12.5308 cm/s, K = 11.6593 cm/s. All the parameters can vary, but due to limited space these can not be discussed here.

5.0 TGA of cotton stalks

The thermo-gravimetric analysis on the fuel sample was conducted in the Thapar University's laboratory. It was conducted to determine the loss in the weight of the fuel sample with temperature. Cotton stalks was grounded to

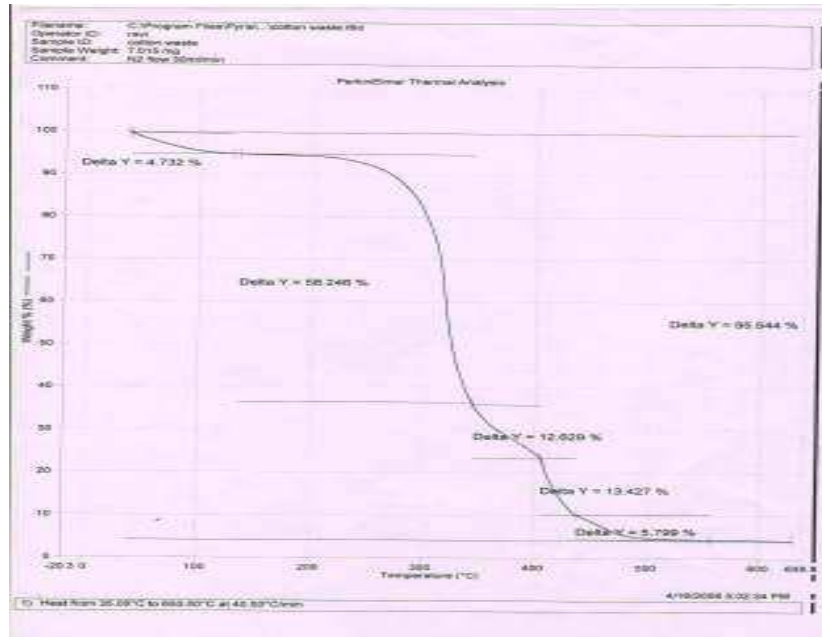


Figure 5. TGA of cotton stalks

powdered form to done the TGA. A curve was obtained for the same which indicated 58.28 % loss in weight at about 350°C. Cotton stalks taken from the power plant has undergone thermo gravimetric analysis at Mechanical Engineering Department, Thapar University using Perkin Elmer TGA. 7.015 g of cotton stalks samples were undergone thermo gravimetric analysis using 100% nitrogen gas. The results from thermo gravimetric analyzer are given in Figure 5. Increasing the temperature from ambient to about 100°C resulted in weight losses of 4-5% due to

loss of water present in the samples and external water bounded by surface tension. The initial decomposition of the cotton stalks started at the temperatures of 220°C (approx.), whereas the final temperatures for the first reaction zone were 340°C (approx.).

Very rapid decomposition of the cotton stalks took place slightly above the initial decomposition temperature at 240°C (approx.), due to the rapid evolution of the volatile products. The average degradation rates observed in this zone were 58.246%. The end of the first reaction zone was accepted as the beginning of the second reaction zone. The total degradation and average degradation rates were lower in the second reaction zone as compared to those of the first reaction zone. The total degradation rate in second reaction zone was 12.629% (approx.) as shown in Figure 5. The second reaction zone ends at approximate temperature of 410°C. The total degradation rate in third and fifth reaction zone is about 13.427 and 5.799 % respectively. Not much change in weight loss is recorded after 520°C.

6.0 Techno economic aspects of above plant

About 73600 tons of cotton stalks are available in area less than 15-25 KM radius. From this 60098 tons is collectable cotton stalks. In less than 15 KM cotton stalks available are 43540 tons of which 37009 tons are collectable, remaining are losses. There are mainly two methods of fuel collection

(a) Through Contractors (b) Direct purchase of biomass fuel by the plant.

(a) Through contractors: Contractors are fulfilling about 70% of the fuel requirement of the plant. Contractors collect the fuel and transport it to plant.

(b) Direct purchase:

Gate purchase from farmers: Farmers bring the fuel directly to the plant through their own mode of transportation. From far of fields: Some biomass is being collected from far off fields by plant itself. Huge cost is involved in collection of surplus cotton stalks available in this region.

7.0 Conclusions

Due to surplus amount of cotton stalks in this region, power plant of 7.5 MW has installed to fulfill the needs of energy. Huge amount of cost is involved in collection of cotton stalks. With the model presented here the various physico chemical parameters been calculated for plant. From the TGA it is clear that all the volatile matter presented could be evolved from the plant up to 560°C. The average size of cotton stalks feed to combustor is 3cm. With the aid of fluidized bed combustion cotton stalks can be usefully converted in to useful energy.

8.0 References

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